Lab: Learning and Model Updating

Learning Goal: Analyze how experience reshapes the generative model.

Part 1: Tracking Parameter Learning

Scenario: You move to a new city. Over 30 days, trace how your generative model updates:

Day	What You Learn	Which Matrix Updates?	How?
Day 1	The bus comes at 8:15 AM	B (transitions)	Learning temporal patterns of transportation
Day 5			
Day 10			
Day 20			
Day 30			

Fill in the table with plausible learning events, identifying which aspect of the generative model (A, B, or D) is being updated and how.

Write your response here

Part 2: Structure Learning in Practice

> **Learning Goal:** Identify cases where simplification (pruning) is adaptive.

Exercise: For each of the following, explain what model simplification occurred and why it was beneficial:

1. A medical student learns to distinguish "important" from "irrelevant" symptoms, focusing only on diagnostic information
2. An experienced driver no longer consciously checks mirrors — it becomes automatic
3. A language learner initially translates word-by-word but eventually thinks directly in the new language

Write your response here

Part 3: The Role of Sleep

> **Learning Goal:** Analyze the relationship between sleep and learning.

Experiment design: You want to test whether sleep helps with learning a new musical instrument.

1. Design a simple experiment: What would you measure? What would the control group do?
2. Based on the distinction between slow-wave sleep (parameter consolidation) and REM sleep (structure learning), what specific predictions does Active Inference make about your results?
3. How would you test whether napping is as effective as a full night's sleep?

Write your response here

Part 4: Skill Acquisition Analysis

Learning Goal: Map the stages of skill learning to prediction error dynamics.

Exercise: Choose a skill you've learned (an instrument, a sport, cooking, a language) and describe your journey through three stages:

1. **Beginner stage**: What prediction errors were most prominent? What was effortful?
2. **Intermediate stage**: What became easier? What was still challenging?
3. **Current level**: What is now automatic? Where do you still make prediction errors?

Relate each stage to the concept of prediction error, precision, and model parameters.

Write your response here

Part 5: Reflection

In 150 words, discuss: Is there such a thing as "too much learning"? When might updating your model too rapidly be harmful? Consider the balance between stability (keeping a good model) and flexibility (adapting to new evidence).

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Temporal analysis	Parameter learning over time
2	Simplification identification	Structure learning / BMR
3	Experimental design	Sleep and memory consolidation
4	Autobiographical analysis	Skill acquisition and prediction error
5	Critical reflection	Stability-flexibility trade-off